

Om Amit Gandhi

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OBJECTIVE

Seeking research and engineering opportunities in distributed systems, blockchain infrastructure, and scalable protocol design. Interested in building simulation frameworks and experimental workflows to evaluate throughput, latency, propagation, and coordination tradeoffs in large-scale decentralized systems.

RESEARCH INTERESTS

Blockchain sharding, distributed systems, scalable consensus protocols, network-aware execution, performance modeling, discrete-event simulation, and reproducible experimental workflows.

EDUCATION

2025 - present Ph.D. in Computer Science at **Illinois Institute of Technology** (GPA: 3.93/4.0)
2024 - 2025 Masters in Computer Science at **Illinois Institute of Technology** (GPA: 3.9/4.0)
2019 - 2023 B.E. in Computer Engineering at **Gujarat Technological University** (GPA: 3.6/4.0)

WORK EXPERIENCE

Research Assistant - Illinois Institute of Technology Aug 2025 - present

- Conduct research in blockchain sharding and distributed systems with emphasis on scalability and coordination tradeoffs.
- Design and develop a discrete-event blockchain simulator to compare sharded and non-sharded architectures under configurable workloads and network conditions.
- Model network-aware block propagation using latency and bandwidth constraints while tracking throughput, confirmation latency, orphan rate, and inflation dynamics.
- Implement protocol mechanisms including transaction pools, fee markets, reward halving, miner/node behavior, and difficulty adjustment.
- Run controlled experiments and parameter sweeps; summarize findings using plots, tables, and reproducible evaluation workflows for ongoing publications.

Teaching Assistant - Illinois Institute of Technology Aug 2025 - May 2026

- Support students through office hours, debugging sessions, and one-on-one guidance across core computer science concepts.
- Evaluate assignments and examinations while providing structured technical feedback to improve student performance.
- Coordinate with instructors and TA teams to standardize grading practices, assignment clarifications, and course communication.
- Identify recurring misconceptions and prepare supporting explanations/resources to improve overall class understanding.
- Assisted Courses: CS458 (Introduction to Information Security)

Junior Web Developer - Minimal Dot

Feb 2023 - Jul 2023

- Developed and maintained client-facing UI components and feature updates using Angular.
- Collaborated with team members on debugging, testing, and iterative improvements across production web applications.
- Improved application usability and responsiveness through front-end optimization and interface refinements.

Web Development Intern - TatvaSoft

Jun 2022 - Jul 2022

- Contributed to the development of a team-based e-commerce bookstore application.
- Implemented role-based user interface workflows aligned with authorization and access-control requirements.
- Worked with TypeScript, PostgreSQL, and .NET fundamentals during application development and testing.
- Participated in collaborative code reviews and proposed workflow improvements during the internship program.

PUBLICATIONS AND MANUSCRIPTS

- Om Amit Gandhi, Ioan Raicu. *Evaluating a Winner-Based Sharded Blockchain Architecture Through Large-Scale Simulation*, under review at IEEE/ACM Big Data 2026.
- Om Amit Gandhi, Ioan Raicu. **Poster Presentation (GCASR 2026):** *Simulation-Based Performance Evaluation of Sharded Blockchain Architectures*.
- Sohini Sahukar, Om Amit Gandhi, Ioan Raicu. **Poster Presentation (GCASR 2026):** *Shards on a Shoestring: Putting Blockchain Sharding Simulations to the Test*.
- Om Amit Gandhi, Ioan Raicu. **arXiv Preprint:** *Simulation-Based Performance Evaluation of Sharded Blockchain Architectures*. [arXiv:2606.14394](https://arxiv.org/abs/2606.14394).
- Samuel Taiwo Fatunmbi, Om Amit Gandhi, Luke Logan. **arXiv Preprint:** *VaultxGPU: GPU-Accelerated Blockchain Consensus*. [arXiv:2606.14007](https://arxiv.org/abs/2606.14007).

PEER REVIEW ACTIVITIES

Conference Reviewer

eScience: IEEE International Conference on e-Science, 2026

SKILLS

Programming	Python, C, C++, TypeScript, JavaScript
Frameworks/Tools	React, Angular, Flask, Node.js, Flutter, SimPy, OpenMP, Docker
Systems/Infrastructure	Linux, Git, distributed systems, blockchain sharding, performance modeling, discrete-event simulation
Research/Writing	L ^A T _E X, Overleaf, experimental evaluation, reproducible workflows
Languages	English, Gujarati, Hindi

VOLUNTEER EXPERIENCE

Volunteer - eScience 2025

Sept 2025

- Served as a volunteer in the eScience conference 2025 held in Chicago.
- Helped with AV support for conference sessions and logistics.

RESEARCH PROJECTS

Blockchain Sharding Simulator (Discrete-Event Simulation)

May 2025 - Present

- Designed and developed a discrete-event blockchain simulator to evaluate scalability tradeoffs between sharded and non-sharded architectures.
- Modeled miners, nodes, transaction pools, and network-aware block propagation using configurable latency and bandwidth constraints.
- Implemented protocol mechanisms including fee markets, reward halving, difficulty adjustment, and workload-driven transaction execution.
- Measured system metrics such as throughput, propagation latency, confirmation latency, orphan rate, and inflation dynamics across varying workloads and network configurations.
- Built reproducible experimental workflows and automated evaluation pipelines for large-scale simulation studies supporting ongoing research publications.
- Code: github.com/OmGandhi091611/Sharding_Simulations

Privacy-preserving Financial Computation using Homomorphic Encryption

Nov 2024

- Built a privacy-preserving computation framework capable of performing financial operations directly on encrypted data.
- Developed a full-stack web application using Python Flask and React to demonstrate encrypted simple-interest and loan-approval workflows.
- Explored privacy-preserving machine learning using Concrete-ML and compared encrypted inference outputs against non-encrypted baselines.

Sentiment Classification using Machine Learning

Apr 2024

- Developed a supervised machine learning pipeline for binary sentiment classification on text datasets.
- Implemented preprocessing and feature-engineering workflows including tokenization, normalization, and vectorization for model training.
- Evaluated classification performance on user-provided real-time inputs and compared prediction behavior across different text samples.

PROJECTS

Personal Portfolio

Jul 2024

- Built a macOS-inspired, window-based portfolio UI featuring draggable/resizable windows, dock interactions, and app-style navigation.
- Implemented mobile-responsive layouts with a full-screen app grid and touch-friendly controls to match phone UX constraints.
- Integrated a unified audio player/control center with play/pause, track controls, progress tracking, and volume control across desktop and mobile views.
- Designed a Finder-like experience to showcase projects and a contact workflow suitable for deployment on Vercel.

Plane 3D animations

Jul 2023

- Built a 3D web experience with scroll-based plane animation.
- Used React Three Fiber / Three.js learning resources to implement WebAR/3D features.

REFERENCES

Prof. Ioan Raicu

Illinois Institute of Technology

Email: iraicu@illinoistech.edu

Associate Professor

Prof. Andre Bauer

Illinois Institute of Technology

Email: abauer7@illinoistech.edu

Assistant Professor

PROFESSIONAL MEMBERSHIPS

Member, Upsilon Pi Epsilon Honor Society (UPE)

2025 - present

- Admitted into Upsilon Pi Epsilon (UPE) for strong academic performance in computer science.
- Engaged with the computing honor society community and academic network.